

Miller Rabin Algorithm

Module I

Miller Rabin algorithm

1. Let $n - 1 = 2^s \cdot d$ where $d \in N$ and $s \in N$.
[choose a random integer a with $2 \leq a \leq n - 2$]

2. Compute

$$X \equiv a^d \pmod{n}$$

If $X \equiv \pm 1 \pmod{n}$ then end the algorithm with message “ n is probably prime”.

If $s = 1$ then end the algorithm with message “ n is definitely not prime”.

Otherwise set $r=1$ and go to step 3.

3. Compute

$$X \equiv a^{2^r d} \pmod{n}$$

If $X \equiv 1 \pmod{n}$ then end the algorithm with message “ n is definitely not prime”.

If $X \equiv -1 \pmod{n}$ then end the algorithm with message “ n is probably prime”.

Otherwise set $r=r+1$ and go to step 4.

4. If $r = s - 1$ then go to step 5.

5. Compute

$$X \equiv a^{2^r d} \pmod{n}$$

If $X \not\equiv -1 \pmod{n}$ then end the algorithm with message “ n is definitely not prime”.

If $X \equiv -1 \pmod{n}$ then end the algorithm with message “ n is probably prime”.

Check if 7 is prime number?

- For $n=7$, $n-1 = 2^s \times d$ (d is odd number)
- $\Rightarrow$ express $n-1$ in $2^s \times d$ then $7-1 = 2^1 \times 3$
- $\Rightarrow 6 = 2^1 \times 3$, $s=1$ and $d=3$.
- Then compute, $X \equiv a^d \pmod{n}$, $2 \leq a \leq n-2$. that means $2 \leq a \leq 5$. select $a=3$.
- $X \equiv 3^3 \pmod{7} \Rightarrow 27 \pmod{7} = 6 \Rightarrow X = 6$.
- Then check, $X \equiv \pm 1 \pmod{n} \Rightarrow 6 \equiv \pm 1 \pmod{7}$
- $6 \equiv 1 \pmod{7}$? $\Rightarrow 6 \pmod{7} \equiv 1 \pmod{7} \Rightarrow 6 \not\equiv 1$.
- Now check $6 \equiv -1 \pmod{7}$? $6 \pmod{7} = -1 \pmod{7} \Rightarrow 6 = (7 - (1 \pmod{7})) \Rightarrow 6 = (7-1) \Rightarrow 6 = 6$. **Then declare “n is probably prime”.**

In the first two steps of the Miller Rabin algorithm we got the answer

Check if 15 is prime number?

- For $n=15$, $n-1 = 2^s \times d$ (d is odd number)
- $\Rightarrow$ express $n-1$ in $2^s \times d$ then $15-1 = 2^1 \times 7$
- $\Rightarrow 14 = 2^1 \times 7$, $s=1$ and $d=7$.
- Then compute, $X \equiv a^d \pmod{n}$, $2 \leq a \leq n-2$. that means $2 \leq a \leq 13$. select $a=3$.
- $X \equiv 3^7 \pmod{15} \Rightarrow 2187 \pmod{15} = 12 \Rightarrow X = 12$.
- Then check, $X \equiv \pm 1 \pmod{n} \Rightarrow 12 \equiv \pm 1 \pmod{15}$
- $12 \equiv 1 \pmod{15}$? $\Rightarrow 12 \pmod{15} \equiv 1 \pmod{15} \Rightarrow 12 \neq 1$.
- Now check $12 \equiv -1 \pmod{15}$? $12 \pmod{15} = -1 \pmod{15} \Rightarrow 12 = (15 - (1 \pmod{15})) \Rightarrow 12 = (15-1) \Rightarrow 12 \neq 14$. **That means $X \not\equiv \pm 1 \pmod{15}$.**
- Now check $s=1$, then **declare “ n is definitely not prime”.**

In the first two steps of the Miller Rabin algorithm we got the answer

Check if 29 is prime number?

- For $n=29$, $n-1 = 2^s \times d$ (d is odd number)
- $\Rightarrow$ express $n-1$ in $2^s \times d$ then $29-1 = 2^2 \times 7$
- $\Rightarrow 28 = 2^2 \times 7$, $s=2$ and $d=7$.
- Then compute, $X \equiv a^d \pmod{n}$, $2 \leq a \leq n-2$. that means $2 \leq a \leq 27$. select $a=3$.
- $X \equiv 3^7 \pmod{29} \Rightarrow 2187 \pmod{29} = 12 \Rightarrow X = 12$.
- Then check, $X \equiv \pm 1 \pmod{n} \Rightarrow 12 \equiv \pm 1 \pmod{29}$
- $12 \equiv 1 \pmod{29}$? $\Rightarrow 12 \pmod{29} \equiv 1 \pmod{29} \Rightarrow 12 \neq 1$.
- Now check $12 \equiv -1 \pmod{29}$? $12 \pmod{29} = -1 \pmod{29} \Rightarrow 29 = (29 - (1 \pmod{29})) \Rightarrow 12 = (29-1) \Rightarrow 12 \neq 28$. **That means $X \not\equiv \pm 1 \pmod{29}$.**
- Now check s , which is $s \neq 1$, then set $r=1$ go to next step.

Check if 29 is prime number? Cntd..

- $x \not\equiv \pm 1 \pmod{29}$ and $s \neq 1$ then set $r=1$ go to next step.
- $s=2, a=3, d=7$.

for($r=1; r \leq s-1; r=r+1$)

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$a^{2^r d} \pmod{n} \equiv 1 \pmod{n}$, then “ n is definitely not prime”.

$a^{2^r d} \pmod{n} \equiv -1 \pmod{n}$, then “ n is probably prime”.

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Check if 29 is prime number? Cntd..

- $X \not\equiv \pm 1 \pmod{29}$ and $s \neq 1$ then set $r=1$ go to next step.

- $s=2, a=3, d=7.$

for($r=1; 1 \leq 1; r=r+1$)

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$$3^{2^{17}} \pmod{29} \equiv +1 \pmod{29} ?$$

$$3^{14} \pmod{29} \Rightarrow 4782969 \pmod{29} \Rightarrow 28$$

$$28 \not\equiv +1 \pmod{29}$$

$$3^{2^{17}} \pmod{29} \equiv -1 \pmod{29} ?$$

$$28 \equiv 29 - (1 \pmod{29})$$

$$28 \equiv 28 \text{ True}$$

Then declare that “ n is probably prime”.

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Check $n=61$?